DATE : 14.04.2026

DT/NT : DT

LESSON : AWS

SUBJECT : IAM

BATCH : B 424-020DE

AWS-DEVOPS

TECHPRO EDUCATION

techproeducation.com +1 (585) 304 29 59

Table of Contents

- Introduction to IAM
- Root User and IAM Identity Center
- IAM Users and Access Methods
- IAM User Groups
- IAM Policies
- IAM Roles and EC2 Hands-On

TECHPRO EDUCATION

Introduction to IAM

TECHPRO EDUCATION

What is IAM?



IAM = Identity & Access Management

Authentication

Prove your identity

- Username + Password + {MFA}
  or
- Access Key + Secret Key
  or
- Access Key + Secret Key + Session Token

Authorization

Permission to access resources

- IAM Policies
  and/or
- Resource Policies

TECHPRO EDUCATION

Introduction to IAM

Categorizing IAM Components

WHO IS ACCESSING AWS?

IAM User

Long-term identity in one AWS account. Use only when necessary.

IAM Group

Groups users for permissions. A group cannot sign in.

WHAT ARE THEY ALLOWED TO DO?

IAM Role

Assumed by trusted principals; provides temporary credentials.

IAM Policies

JSON rules: actions, resources, effects and optional conditions.

Identity Center: preferred workforce sign-in • Least privilege: grant only required actions

TECHPRO EDUCATION

Introduction to IAM

Modern Human Access — IAM Identity Center

Identity Center

Centralized workforce sign-in

Permission sets -> AWS account roles

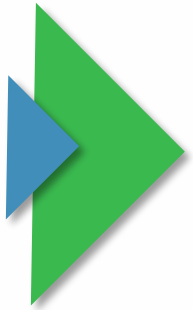
IAM User

Long-term credentials in one account

Use when federation is not suitable

Best practice: require MFA and use short-term, least-privilege credentials.

TECHPRO EDUCATION



IAM Users

TECHPRO EDUCATION

IAM Users

What is IAM User?

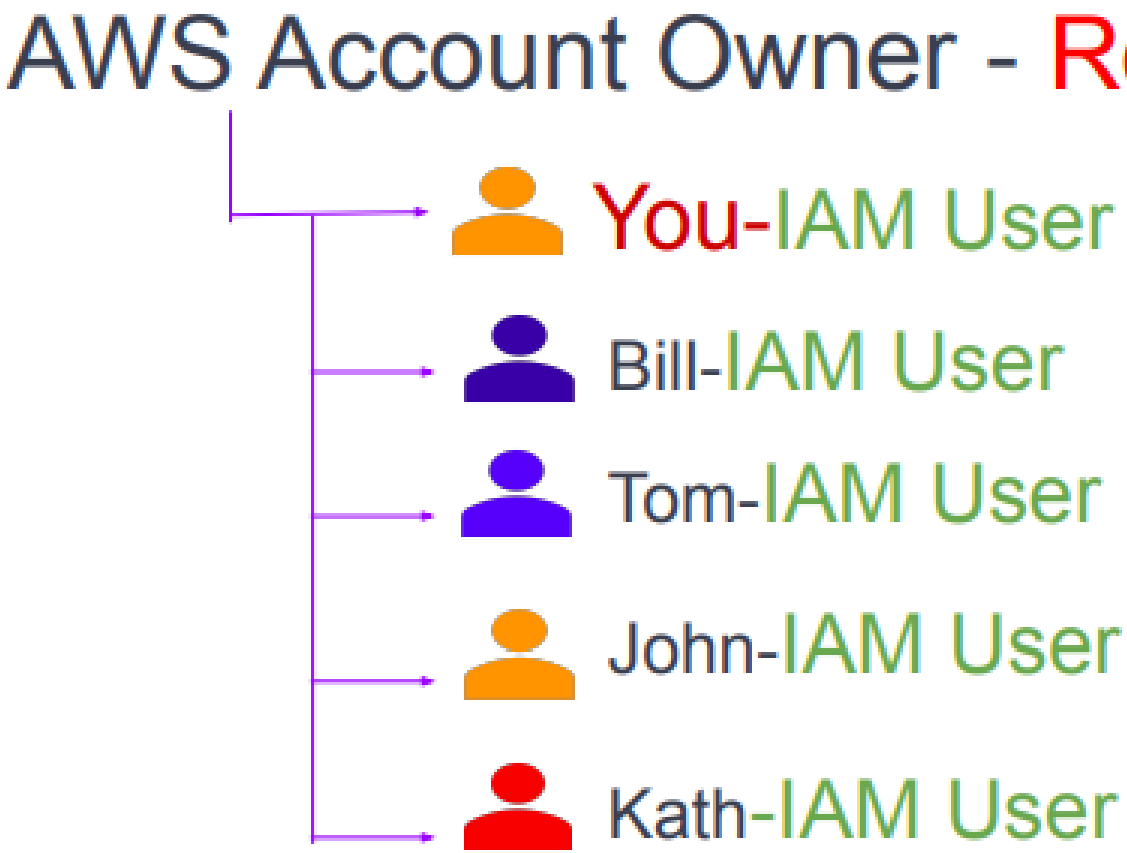


- An IAM user is an identity in one AWS account with permissions and optional long-term credentials. Prefer federation and temporary access for people.

TECHPRO EDUCATION

IAM Users

What is Root User and IAM User.



- Root User is a special user
- Username is email used to create account
- Generally, cannot limit permissions of Root User
- Cannot delete Root User
- Best practices:
  - Enable MFA for Root User
  - Don't user Root User for day-to-day work
  - Keep password in a secure location

TECHPRO EDUCATION

IAM Users

What is Root User and IAM User.

ROOT USER

Account owner credentials

Required for limited account-level tasks

MFA required

Not for day-to-day use

DAILY HUMAN ACCESS

Recommended: IAM Identity Center

Short-lived role-based access

MFA and least privilege

IAM users only when justified

IAM User ≠ Root User | IAM Role ≠ Permanent Access Key

TECHPRO EDUCATION

IAM Users

Access Types

AWS Console

Interactive browser access.

Recommended for people: IAM Identity Center + MFA.

AWS CLI / SDK

Programmatic access from computers.

Prefer short-lived session credentials.

Workload Role

EC2 / Lambda applications receive

temporary credentials automatically.

Long-term access keys are not a default setup step.

TECHPRO EDUCATION

IAM Users

Sign in with Root User- AWS Management Console Access

1. Open AWS sign-in

Choose Root user only when the task genuinely requires the root identity.

2. Authenticate

Enter the AWS account email, password, and required MFA factor.

3. Finish and sign out

Perform the account-only action; sign out when finished.

Never create root access keys for the IAM classroom lab.

TECHPRO EDUCATION

IAM Users

Sign in with IAM User- AWS Management Console Access

IAM USER (TRAINING)

Use the account-specific sign-in URL or account alias, IAM username, password and MFA.

IAM IDENTITY CENTER (PREFERRED)

Use the organization access portal; sign in with configured MFA and select your AWS account/role.

Do not share passwords or access keys between students.

TECHPRO EDUCATION

IAM Users

Access Types

Recommended credentials by use case

Human in terminal

AWS CLI v2 + IAM Identity Center

aws configure sso

EC2 or AWS service

IAM role attached to the workload

Auto-rotating temporary credentials

Legacy exception

IAM access key may be required

Secure, rotate and remove after use

Access key ID + secret access key are sensitive, long-term credentials.

TECHPRO EDUCATION

IAM Users

Sign in with IAM User- Programmatic Access

How applications authenticate securely

EC2 Application

Runs with an attached IAM role.

Temporary Credentials

AWS supplies and refreshes role credentials.

AWS API / S3

SDK or CLI signs authorized requests.

Do not hardcode access keys in source code, scripts or EC2 user data.

TECHPRO EDUCATION

IAM Users

Sign in with IAM User- Programmatic Access

AWS CLI — Preferred Setup for Human Users

1. Configure SSO

aws configure sso

Choose portal / Region / role / profile.

2. Sign In and Verify

aws sso login --profile lab

aws sts get-caller-identity --profile lab

Profile name “lab” is an example; use the name created by the wizard.

TECHPRO EDUCATION

IAM Users

Sign in with IAM User- AWS CLI

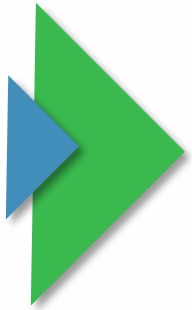
Check who the AWS CLI is using

In your own terminal

```
aws --version
aws configure sso
aws sso login --profile lab
aws sts get-caller-identity --profile lab
```

Expected: Account ID and ARN for your authenticated session.

TECHPRO EDUCATION

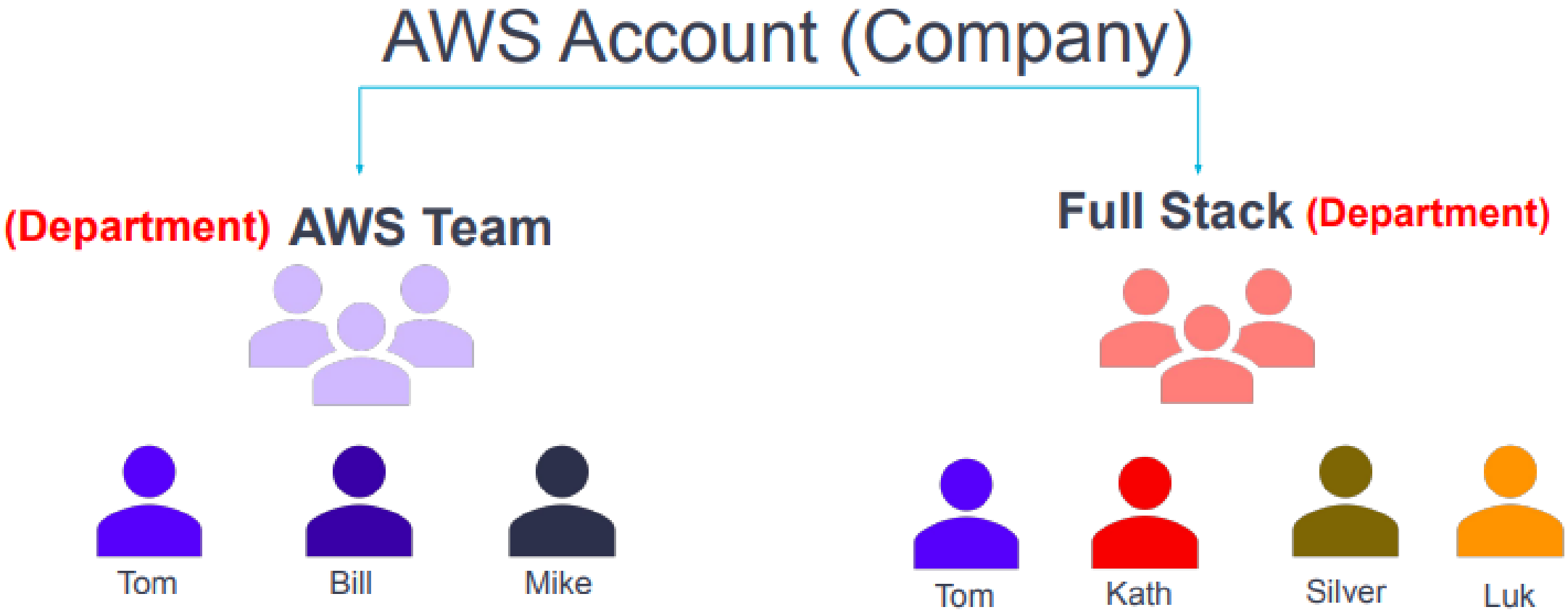


IAM User Groups

TECHPRO EDUCATION

IAM User Groups

What is User Group in AWS?



TECHPRO EDUCATION

IAM User Groups

IAM User Group Features

Policies

Attach managed policies or an inline policy to a group.

Membership

A user can be in multiple groups; permissions combine.

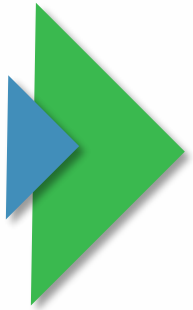
Authentication

Groups do not authenticate and cannot be a Principal.

Quota

Confirm current AWS IAM group/user quotas in your account.

TECHPRO EDUCATION



IAM Policies

TECHPRO EDUCATION

IAM Policies

What is a Policy?

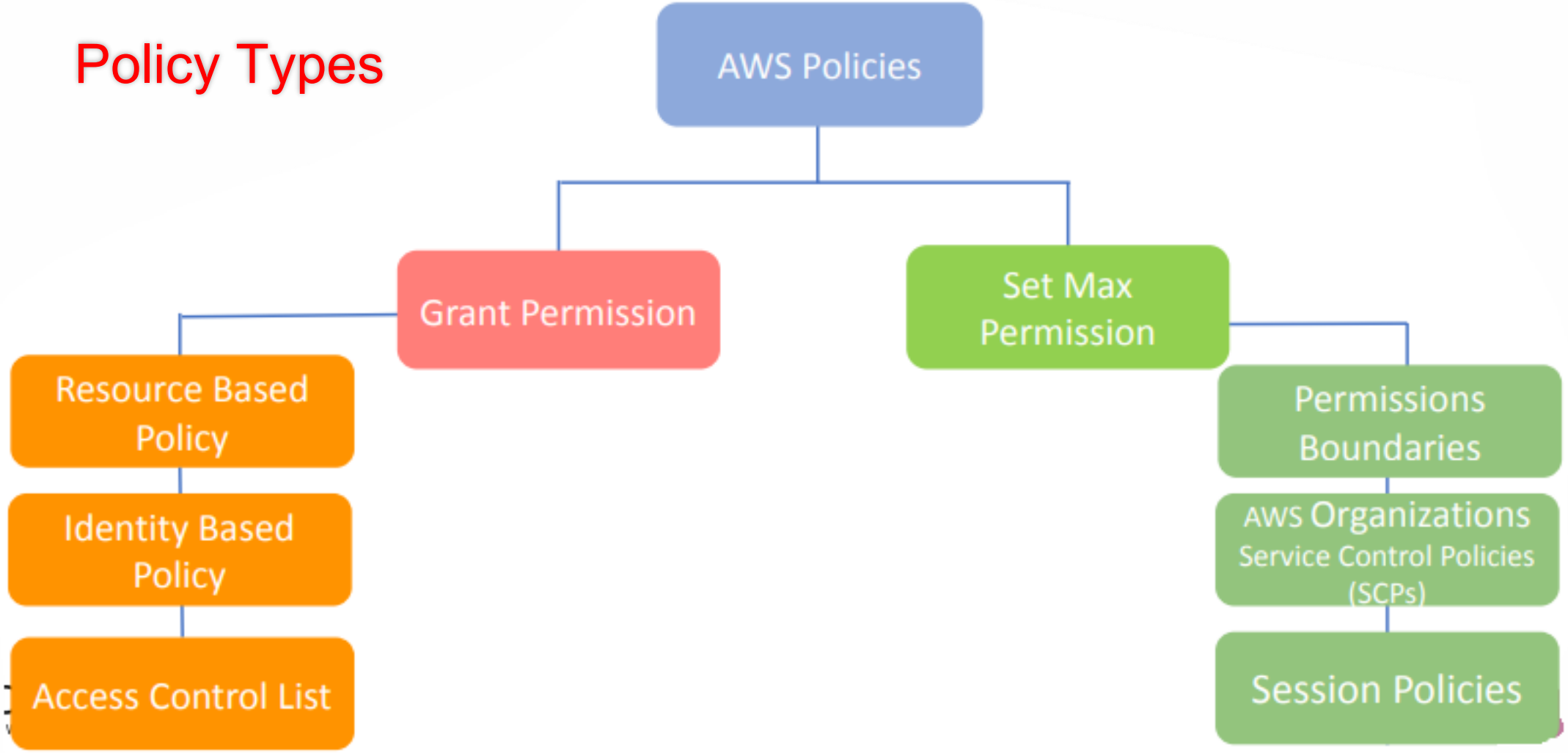


- A policy is a JSON document that defines permissions.
- By default, requests are denied unless allowed.
- An applicable explicit Deny overrides an Allow.
- Other policy types can also limit the final permissions.

TECHPRO EDUCATION

IAM Policies

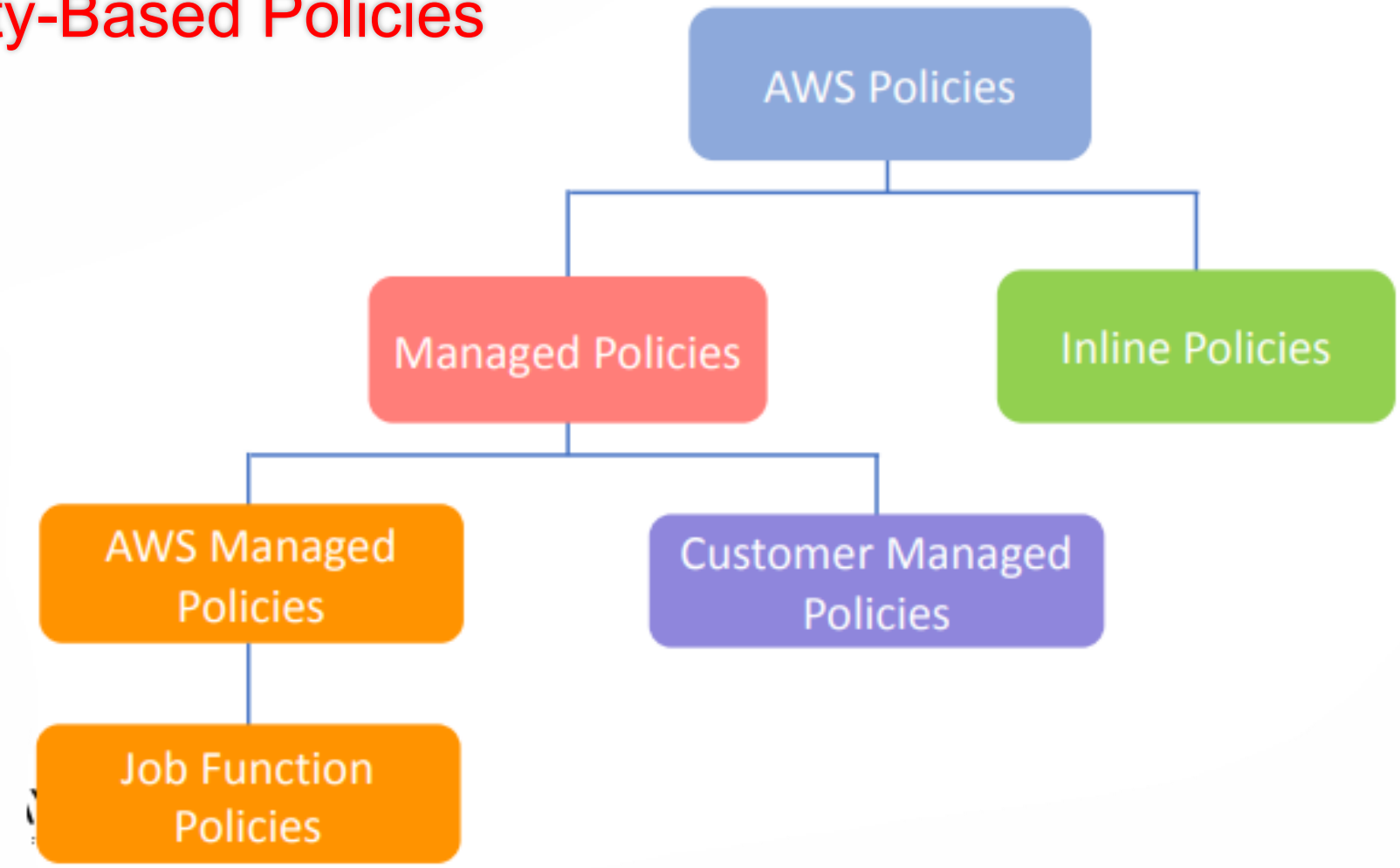
Policy Types



TECHPRO EDUCATION

IAM Policies

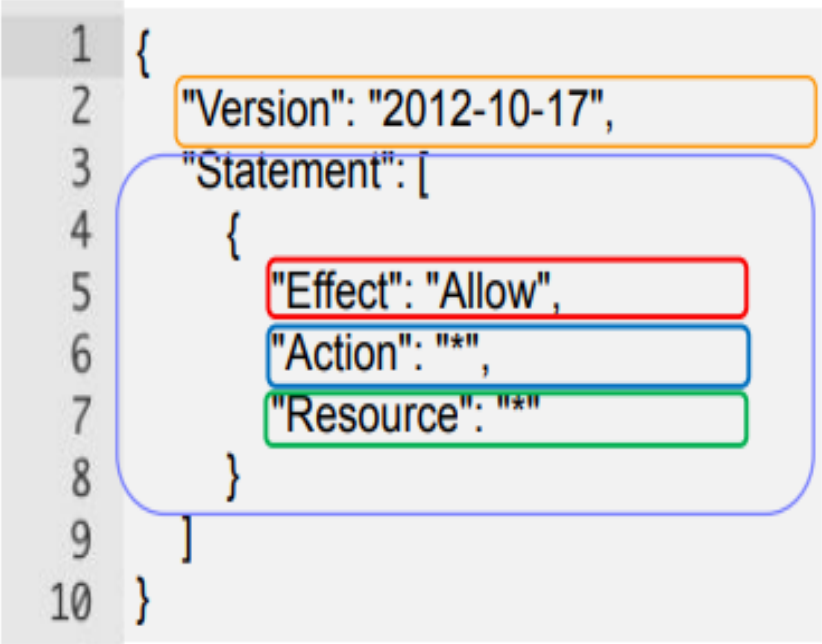
Identity-Based Policies



TECHPRO EDUCATION

IAM Policies

Policies - JSON Identifiers



Version — IAM policy language version (2012-10-17).

Statement — permission rules.

Effect — Allow or Deny.

Action — AWS operation, e.g. s3:GetObject.

Resource — ARN of the applicable resource.

Condition — optional requirement for the rule.

Principal — resource-based or trust policy only.

TECHPRO EDUCATION

IAM Policies

Policy Evaluation — Allow vs. Deny

Default: DENY

Nothing is granted unless an applicable policy allows it.

Explicit ALLOW

Permits an action when other controls do not block it.

Explicit DENY

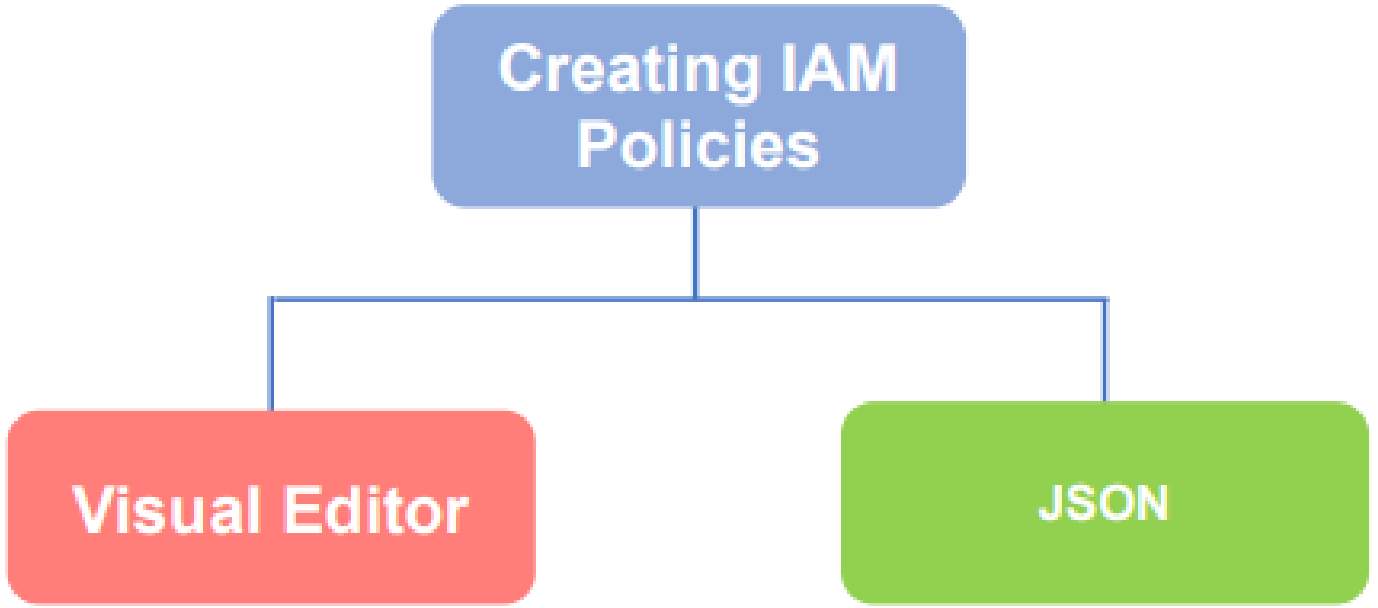
Wins over any Allow in applicable policies.

Permission boundaries, session policies and Organizations policies can also limit access.

TECHPRO EDUCATION

IAM Policies

Creating IAM Policies



IAM Policies

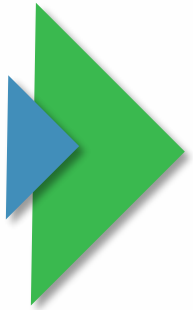
Creating IAM Policies

Example — Read objects from ONE S3 bucket

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::YOUR-LAB-BUCKET/*"
  }]
}
```

Replace YOUR-LAB-BUCKET with your actual private bucket name.

TECHPRO EDUCATION



IAM Roles

TECHPRO EDUCATION

IAM Roles

What is a Role in AWS?



- An IAM role is an identity with a trust policy and permissions.
- Trusted principals can assume the role when permitted.
- Role sessions use temporary security credentials.
- EC2 applications should use roles instead of embedded keys.

TECHPRO EDUCATION

IAM Roles

Who can assume an IAM Role?

People / Federated Users

IAM users, SSO/federated principals when trusted and authorized.

AWS Services

EC2, Lambda, ECS and others — depending on the role trust policy.

Other AWS Accounts

Trusted users/roles in another account with appropriate permissions.

Not IAM Groups

Groups can grant sts:AssumeRole permissions to their users, but cannot assume a role themselves.

TECHPRO EDUCATION

IAM Roles

Two Policies of an IAM Role

Trust Policy

WHO can assume the role?

Example Principal: ec2.amazonaws.com

Action: sts:AssumeRole

Permissions Policy

WHAT can the role do?

Example Action: s3:GetObject

Resource: one bucket/object ARN

Both trust and permissions must fit the actual workload.

TECHPRO EDUCATION

IAM Roles

EC2 reads S3 using a role — no saved access keys

EC2 Instance

Runs AWS CLI / SDK

EC2 instance profile

IAM Role

Trust: ec2.amazonaws.com

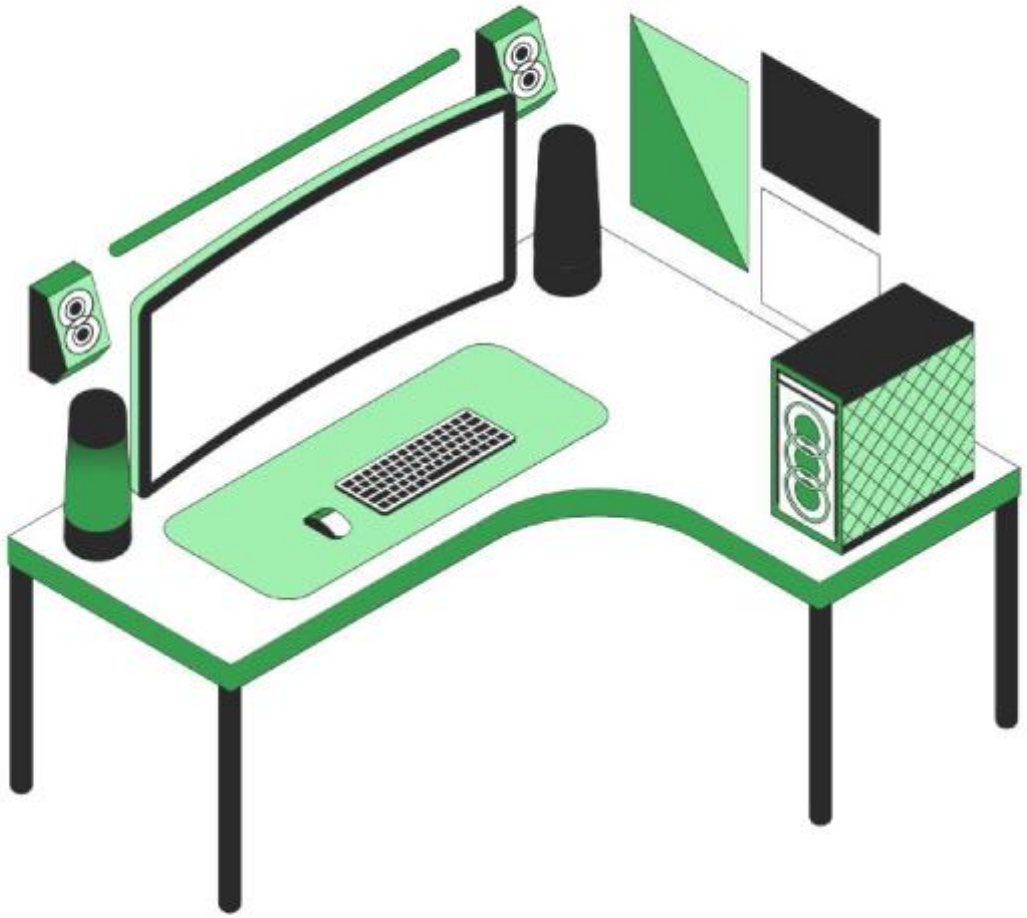
Permissions: S3 GetObject

Private S3 Bucket

Access only to the approved objects.

Instance role → temporary credentials → authorized AWS API request

TECHPRO EDUCATION



Do you have any questions?

Send it to us! We hope you learned something new.

TECHPRO EDUCATION